

# Still Ticking When the Signal Isn't

A Field Validation of a Suburban Bus Stop's Live-Arrival Display Against the Bus's Own GPS Trace

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## Motivation

Live-arrival signs promise a countdown tied to a bus's real position. Operators rarely disclose what a sign shows once that position stops arriving. Riders have no way to tell a genuinely tracked minute from a timetabled guess wearing the same countdown format.

## Research questions

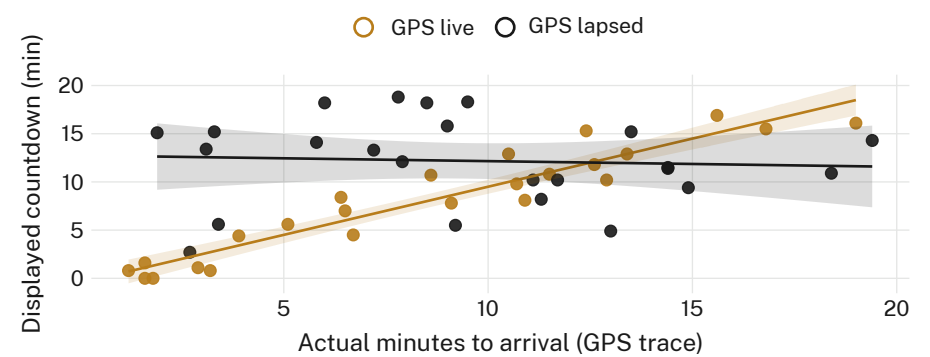
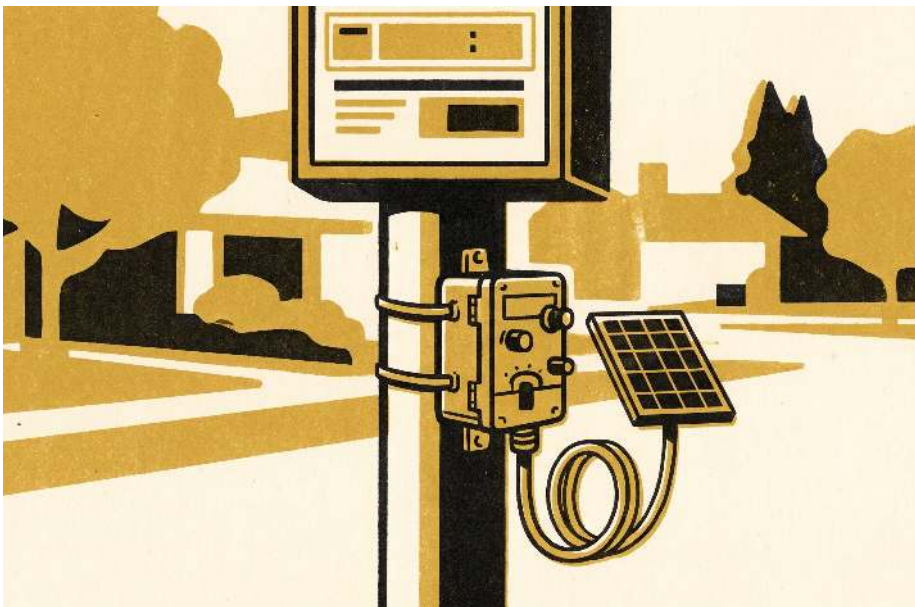
- Does the displayed countdown track the bus's GPS-derived arrival time?
- Does that agreement hold once the vehicle's GPS feed drops out?
- Can a rider tell a tracked minute from a timetabled one from the display alone?

## Study design

- 1 suburban stop · one route · 15-week deployment; solar logger photographed the sign every 30 s
- 48 arrivals matched to the operator's public GPS feed, spanning 19 recorded dropouts
- display format (icon, font, update cadence) held constant across both states — pre-registered check

## What we found

The display tracks the bus closely while its GPS feed is live ( $r = 0.95$ ) and barely at all once it lapses ( $r = -0.06$ ,  $n = 48$ ). Nothing about the sign changes to say which is happening.



Displayed-vs-actual agreement collapses from  $r = 0.95$  to  $r = -0.06$  across 19 GPS dropouts logged between February and May 2026 — the sign's icon, font, and update cadence never varied.

## Implications & where next

A rider has no way to tell a genuinely tracked minute from a timetabled guess in the same font. The sign's confidence is manufactured, not measured, exactly when the position it claims to show has gone missing, and the dropout is disclosed nowhere a rider would think to look for it. Next: an on-sign fallback indicator, tested against operator appetite for confessing the gap, and a longer deployment across a second, busier route to see whether dropout frequency scales with vehicle density.

## Selected references

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3. Kim, H.-I., & Park, K.-D. (2025). Satellite positioning accuracy improvement in urban canyons through a new weight model utilizing GPS signal strength variability. *Sensors*, 25(15), 4678. doi:10.3390/s25154678
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We thank the residents of the instrumented street for tolerating the pole-mounted logger, and the operator's control room for the read-only feed access.

No commuter faces retained; only the sign's own display was logged.

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